

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- | | | |
|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ | The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ | A description of all covariates tested |
| ☐ | ☒ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	The DIA MS data were analyzed using DIA-NN 1.8.1. The Skyline(3.6.0.1) was used for quantifying PRM-MS data.
Data analysis	Cytoscape software(3.9.1) were used for the visualization of PPI network. The R(4.4.1) software was used for data analysis. The Python(3.13.1) was used for machine learning model selection and variable selection for inclusion in the model.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The Proteomics data that support the findings of this study have been deposited in IPROX with the PXD067174 accession codes. Source data are provided in this paper.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

The study applies to one sex but not to gender identity. Sex and gender identity were not considered in the study design. This study collected information related to sex, with 75% male and 25% female in the discovery cohort, and 70% male and 30% female in the validation cohort.

Reporting on race, ethnicity, or other socially relevant groupings

In our study, the socially relevant categorization variable used was sex (male and female). This variable was chosen because it was considered an important demographic factor in relation to the study objectives. We explicitly note that sex was not used as a proxy for other variables such as gender identity or socioeconomic status, as these are separate and distinct factors. The term sex was defined as a biological attribute (male or female) and was provided by the participants during data collection through self-reporting. Participants were asked to identify their sex, which was then classified accordingly. To control for confounding variables, we ensured that other demographic factors (e.g., age, pre-existing conditions) were taken into account in the analysis. We used statistical methods to adjust for these variables, ensuring that any observed effects were not due to confounding influences.

Population characteristics

See above

Recruitment

The recruitment of patients took place between June 2024 and April 2025, involving patients who met the inclusion and exclusion criteria. To address potential selection bias, we matched participants using age and sex as covariates in a 1:2 ratio.

Ethics oversight

This study was approved by the Ethics Committees of Beijing Ditan Hospital, Capital Medical University (DTECKY2023-045-01), Huashan Hospital, Fudan University (KY2020-679), and the First Affiliated Hospital, Zhejiang University (IIT0230519B-R1).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Sample size calculation was not performed. However, we grouped patients based on whether they developed secondary bacterial infections or not, in a 1:2 ratio.

Data exclusions

No data was excluded from the analysis.

Replication

No replication measurement of samples was performed in the study.

Randomization

The discovery group and validation group in this study were two independent cohorts, without random grouping.

Blinding

The study does not involve blinding during data analysis. This study is observational, with no patient interventions, and it observes whether patients experience secondary bacterial infections during hospitalization.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involvement in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involvement in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	LYZ (ab267798), DPT (ab272206), CALM1 (catalog no. SEG095Hu) and SERPIND1 (ab277402)
Validation	All primary antibodies were validated for the relevant species and applications based on manufacturer-provided validation and supported by data generated in this study. LYZ (ab267798), DPT (ab272206), and SERPIND1 (ab277402) are Abcam antibodies with application- and species-specific validation statements available on the manufacturer's website. CALM1 (SEG095Hu) was supplied with manufacturer validation for human samples. In this study, antibody specificity was further supported by protein-level validation.

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A